

Mobile App — Improvement Tasks

Wergonic Flutter App

June 2026

Phase 1: Terminology Alignment

Rename UI labels across all 5 translation files (en, de, nl, sv, es) and any hardcoded Dart strings. These are display-only changes — do not rename database fields, API payloads, or model classes.

Current	New	Where
“Work Task” (in free sessions)	“Activity”	Setup wizard, recording screen, session history, anywhere outside Work Cycles
“Work Label”	“Condition”	Add label screen, session metadata, history
“Work Break” / “breakSession”	“Break”	Recording screen, session history
“Virtual Coach” (settings section)	“Feedback Settings”	Settings screen section header
“Finish measurement”	“End Measurement”	Recording screen button
“Leave measurement”	“Discard & Exit”	Recording screen button
“Instant feedback refresh rate”	“Feedback interval”	Settings

No change needed: “Measurements” (list title), “New measurement” (home), “Work Task” inside Work Cycles — these are already correct.

Phase 1b: Backend APIs (required for Phase 2)

The mobile app needs these endpoints to support conditions and activities. The data structures already exist in the web prototype.

- **Activity list endpoint** — fetch the org’s activity groups and their activities (closed list the user picks from during recording)
- **Condition groups endpoint** — fetch the org’s condition groups with selection type (single-choice / multi-choice) and their values
- **Extend sync payload** — the measurement sync must include the selected condition values so they are stored on the backend

Phase 2: Measurement Setup Flow Redesign

2.1 Home Screen

Currently the home screen is nearly empty — just a FAB and sensor status. 80% of app opens are repeat setups. The home should make repeating fast.

- **Quick Start card** — shows the last measurement setup. If last was a Work Cycle: “Record [Worker] at [Station]?”. If last was free: “Record [Worker]?”. One tap reconnects sensors, recalibrates, and starts recording.

- **Sync banner** — “3 measurements waiting to sync” with a sync button. Only visible when there are pending measurements.
- **Recent measurements** — last 3–5 as compact cards (worker name, date, duration, sync status). Tap to view or sync.
- **Sensor status row** — battery levels and connection state for cached sensors. Always visible so the ergonomist is never surprised by a dead sensor.

2.2 Worker Selection Step

Currently just a search field. Most ergonomists work with the same 5–10 workers repeatedly.

- **Recent workers** — show the last 5 used workers as avatar chips at the top. One tap to select.
- Search bar below for other workers
- “New Worker” button at the bottom

2.3 Conditions Step (NEW)

Conditions are whole-measurement metadata (e.g., Shift Type, Environment). They are defined per org in the Work Library. The mobile app currently has no support for conditions.

- Add a new step after worker selection, before work setup
- Show each of the org’s condition groups as a visual selector:
 - **Single-choice groups** (e.g., Shift Type) → horizontal chip row — tap one
 - **Multi-choice groups** (e.g., Environment) → toggleable chips — tap to select multiple
- If the org has no conditions configured, skip this step automatically
- Ask at the start of the measurement, not the end — conditions are known upfront (shift, weather) and asking at the end risks being skipped

2.4 Work Setup Step

Currently the user picks between “Work Task” and “Work Cycle” — confusing terminology and forced choice. This step should adapt to what the org actually has configured.

- If org has **only Work Cycles** → go straight to the hierarchy picker (Factory → Line → Station → Operator → Model). Show recently used cycles at the top.
- If org has **only Activities** (no Work Cycles) → show the org’s activity list. Let the user check which activities they expect to record. These become quick-switch buttons during recording.
- If org has **both** → show a simple choice: “Follow a Work Cycle” or “Free Measurement.”
- This replaces the current “Work Task / Work Cycle” toggle with something that makes sense to the user.

2.5 Pairing Improvements

- If sensors are cached from last measurement, auto-reconnect and show progress per sensor
- Limb-specific status: “Reconnecting Left Arm...” instead of generic “Connecting...”
- Show battery level and signal strength per sensor during pairing
- If all sensors reconnect successfully, **auto-advance to calibration** without requiring the user to tap Next

2.6 Calibration Improvements

The current calibration flow with videos works well. Only small additions:

- Add step progress: “Calibrating 2 of 3”
- After the last calibration succeeds, **auto-advance to recording** — one less tap

Phase 3: Recording Screen Overhaul

3.1 Risk-Colored Limb Tiles

Currently the tiles show raw numbers (32°, 45°/s) which require expertise to interpret. The user needs to know one thing: is this good or bad?

- Change tile backgrounds to **green / yellow / red** based on standard ergonomic angle thresholds (ISO 11226, EN 1005-4 — the same ranges referenced by RAMP, RULA, and REBA)
- Example: arm elevation <30° = green, 30–60° = yellow, >60° = red
- These thresholds are universal across all assessment methods and safe to apply for any org
- Keep the numbers visible but make them secondary — the color is the primary information

3.2 Top Bar: Current Activity + Time

- Show which activity/task is being recorded and elapsed time
- Work Cycle: “Task 3/7: Quality Check · 02:15”
- Free: “Lifting · 12:34”

3.3 Work Cycle: Task Timeline Bar

The current auto/manual toggle works well. These additions give the user more context and control:

- Horizontal progress bar below the tiles showing all tasks as proportional segments (completed = solid, current = highlighted, upcoming = dimmed)
- **Auto mode:** show countdown for current task (“Next in 0:45”). Add an “Extend 30s” button when nearing zero. Show a brief notification when task advances: “Now: [Next Task]”
- **Manual mode:** show elapsed vs expected time as a reference (“2:15 / 3:00”). Make the “Next Task” button bigger and more prominent.

3.4 Free Session: Activity Bar

- Show current activity as a chip at the bottom of the recording screen
- “Switch” button → shows the activities pre-selected in the setup step as quick-switch buttons
- “+” button → full activity list for adding one not pre-selected
- Starting a new activity automatically ends the previous one — no extra tap needed

3.5 Break Behavior

- Adding a break should auto-pause the measurement
- Ending the break should auto-resume
- Breaks appear as distinct segments in the task timeline

Phase 4: Completion + Measurements List

4.1 Completion Screen

Currently the summary after recording is minimal. The user has no confirmation that all metadata was captured.

- After recording ends, show a **completion checklist**: worker ✓, duration ✓, limbs recorded ✓, activities ✓, conditions ✓
- If conditions were not set during setup, prompt here: “No conditions set → Add now”
- “Sync Now” as primary action, “Save & Sync Later” as secondary

4.2 Measurements List

- Add filter chips above the list: All / Synced / Pending / Today / This Week
- Richer cards: worker name, date, duration, activity count, sync status
- Better empty state: illustration + “No measurements yet” + CTA button
- Swipe left to delete

Phase 5: Settings + Onboarding + Work Library

5.1 Settings Restructure

Currently all settings are in one long scrollable page with technical labels.

- Reorganize into groups: **Sensors & Hardware**, **Feedback During Recording**, **Measurement Defaults**, **Organization**, **Work Library** (read-only), **App**
- Add short descriptions under each section header (e.g., “Control how the app alerts you during recording”)
- Hide advanced/developer settings completely for normal users

5.2 Work Library (Read-Only)

- Add a section in settings where the ergonomist can view the org's Work Library:
 - Activities — activity groups and their activities
 - Conditions — condition groups and their values
 - Work Cycles — the hierarchy (Factory → Line → Station...)
- Read-only — editing happens on the web

5.3 Onboarding

- 3 screens for first-time users:
 - Screen 1: “Wergonic measures how your workers move”
 - Screen 2: “Set up sensors, record a measurement, get risk insights”
 - Screen 3: “Let's start your first measurement” → straight into the setup wizard

Do Not Change

- **BLE / sensor layer** — just stabilized after the native Kotlin migration
- **Sync mechanism** — works, don't refactor
- **Calibration logic and videos** — working well, only add UI around them
- **Database fields / API payloads** — all renames are UI-only, backend migration is separate
- **Dark theme** — keep it, works for industrial environments
- **“Work Task” inside Work Cycles** — stays as is